Data471 Final Project

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Executive Summary

Difficulty accessing food due to economic or geographic barriers can significantly impact a person's quality of life and exacerbate existing inequalities. This report examines how the relationship between demographic composition and low income, low access status varies across ethnic groups, and between Urban and Rural Census Tracts in the United States.

The analysis uses the 2019 Large Retailer Access Map (LRAM), based on 2010 Census data. Alaska and Hawaii were excluded due to their geographic separation and historical distinctions. Census Tracts with populations below 150 were also excluded. The outcome measures whether a tract is classified (1) or not classified (0) as Low Income and Low Access, using access thresholds of 1/2 mile in Urban areas and 10 miles in Rural areas. All ethnic groups available in the dataset (represented as percentages of the Census Tract's total population) were analysed, along with differences between Urban and Rural areas.

A model using data on an adjusted natural **logarithmic scale** provided the best fit. Consequently, differences at larger magnitudes have a smaller impact than differences at smaller magnitudes; an increase from 1% to 2% has a greater impact than an increase from 10% to 11%.

A higher percentage of Black individuals in a Census Tract indicated greater odds of being Low Income and Low Access, where an increase from (50% increase per % increase on logarithmic scale, 95% CI (1.48, 1.52), p-value $<$ 2e-16) for both Urban and Rural areas.

A similar increase in odds was discovered for increasing percentages of **Native Americans/Alaska Natives** in Urban areas (51% increase per unit increase on the logarithmic scale, 95% CI (1.42, 1.60), p-value $<$ 2e-16) and even larger for Rural areas at a near 100% increase per unit on the logarithmic scale (additional multiplicative effect of 32% for Rural Census Tracts, 95% CI (1.21, 1.45), p-value = 6.32e-10). For **Hawaiian/Other Pacific Islanders** the odds increase by 115% in both Urban and Rural areas (95% CI (1.93, 2.39), p-value $<$ 2e-16).

Different relationships were found in Urban and Rural areas for the percentage of **Hispanic** people: a decrease in odds was observed in Urban Census Tracts with more Hispanic people, and an increase in odds was observed in Rural Census Tracts with more Hispanic people. More analysis, either qualitative or quantitative, may be done to better understand the impact on this ethnic group.

Targeted initiatives should prioritise or strongly consider the impact on areas with a stronger presence of Black, Native American, Alaskan Native, Hawaiian, and other Pacific Islander groups, to prevent exacerbating existing inequalities these communities face. Initiatives that increase food access in areas with a higher

percentage of these demographics may have a greater impact. The potential impacts on Urban and Rural Census Tracts should generally be considered separately. Further research could also explore the relationship between the number of children or elderly within specific ethnic groups, with low income and/or low access.

Background

Reasonable access to food sources has a significant impact on a person's quality of life. One key component of accessibility is the distance a person is required to travel in order to access affordable and healthy food, such as ingredients from a supermarket. This is a commonly discussed phenomenon in the United States of America, and is referred to as "food deserts".

This impact of longer distances is exacerbated for those with less financial resources, such as individuals in poverty. These individuals have less resources to overcome the geographical barrier to access, such as the money required for extra fuel, or the ability to accumulate savings to buy in bulk.

The Food Access Research Atlas (FARA) provides various measures of food access in the United States by Census Tract, in combination with data regarding income, poverty rates, age and ethnicity breakdowns. Census Tracts are geographical regions defined such that they roughly equally divide the population.

The aim of this analysis is to understand in greater detail the characteristics of Census Tracts that are considered to have a significant number of individuals that both suffer from low income and low access to food.

Specifically, I aim to answer the question:

> *How do the **demographic characteristics** of a Census Tract relate to whether it is considered to be an area that suffers from **low income** and **low access**?*

Addressing this question can achieve the following:

1. It may highlight unforeseen consequences, or potentially hidden biases, in decisions made by those that influence the location of supermarkets, such as local governments and supermarket businesses
2. It may identify areas and population groups that future initiatives should prioritise, to have the greatest impact

The FARA dataset and its use in this report

We used the 2019 Large Retailer Access Map (LRAM) data, part of the United States Department of Agriculture's Food Access Research Atlas (FARA, USDA ERS 2019 – see References). This data is available directly from the USDA ERS LRAM data download.

This data describes, by Census Tract, the accessibility of supermarkets to the population of that Census Tract. Food Access is defined in multiple variables at several distances - for example, **LAPOP1_10** describes the number of people that are more than 1 mile from a supermarket for an Urban Census Tract, and more than 10 miles for a Rural Census Tract. Demographic information such as the population of different ethnicities, the population of children and seniors, is available, both as totals for the census and as the total of each demographic that is more than a specific distance from a food vendor, as measured in the 2010 Census. All 50 US states are covered.

Data types and structure

The data contains one row per census tract, with 72,531 rows. Each row also specifies the State and County (character/categorical). The rest of the data exhibits three variable types: binary flags (e.g. **Urban** flags if the Census Tract is an Urban area, **LATracts1** flags if the tract is considered to be low access at a 1 mile distance); tract totals (numeric counts, e.g. **POP2010** for the total population of the tract, **TractKids** for the number of children in the tract); and continuous rates/shares (continuous, e.g. **lapophalfshare** details the share of the tract that are beyond 1/2 mile from a supermarket).

Each population count that represents individuals with insufficient food access, low income, or both, has a corresponding flag. For example, **LAPOP1_10** counts the population of individuals who are more than 1 mile from a supermarket in urban Census Tracts and 10 miles in rural Census Tracts. In comparison, **LA1and10** is a flag to indicate where either at least 500 people, or at least 33%, of the population of that Census Tract are more than 1 mile from a supermarket in urban Census Tracts and 10 miles in rural Census Tracts.

Variables used in this analysis

The following variables were used in this analysis:

- Flags representing if a Census Tract is considered to be **Low Income**, **Low Access**, and **combined Low Income and Low Access**, where Low Access is measured by a threshold population that is more than 1/2 mile or 10 miles from a supermarket for Urban and Rural areas respectively. Definitions of both low income and low access can be found in the LRAM Reference Guide (2)
- **Urban / Rural**, a flag denoting the population density type of the Census Tract. Urban is the reference level (Urban = 0, Rural = 1)
- The total **population** of the Census Tract
- Population demographics for the Census Tract as percentages, including:
  - **Kids** (under 18)
  - **Seniors** (65 or older)
  - **White**
  - **Black**
  - **Asian**
  - **Hispanic**
  - **Hawaiian/Other Pacific Islander**
  - **Native American/Alaska Native**
  - **Other races/Multiracial**
- **Median Family Income** in the Census Tract, in USD
- **Poverty Rate**: the percent of the population "living with income at or below the Federal poverty thresholds by family size" (2)

Completeness; handling missing values in EDA and Detailed Analysis

The following columns were fully populated:

- State, County, Urban flag (basic identifier attributes)
- **Pop2010** (total population count of the census tract)
- All flag variables, such as **LA1and10** (flag for low access at 1 mile and 10 miles, urban and rural respectively)

Population tract totals (for example, **TractBlack**, the total number of Black people in the population), Median Family Income, and Poverty Rate, are missing for only 4 records.

For the Exploratory Data Analysis, missing Median Family Income and Poverty Rates are imputed using their respective medians across all Census Tracts, while missing demographic population counts (for example, **TractBlack**) are replaced with zeros. For the Detailed Analysis, the 4 records with missing data are removed.

Total population counts for low access exhibited many more missing values - between 20% and 50%, depending on the distance measure. Filled records for these fields included measurements of 0, indicating a missing value is conceptually different to the measurement equalling 0. Corresponding flags are zero when the population count is not available. Variables for low access and low income shared the same percentage of missing values as their low access only counterparts.

No clear pattern was found among the missing values. We suspect that these values may have been left blank if the Census Tract was smaller than the distance described in the variable.

For both the Exploratory Data Analysis and Detailed Analysis, Low Access and Low Income, Low Access flags measured at 1/2 and 10 miles were analysed. No records were removed or changed where the corresponding Low Access and/or Low Income, Low Access population counts were missing. Including these records prevents any potential bias (20.05% of records would be affected), and assumes that the decision to impute the corresponding flags as 0 was deliberate and appropriate.

Ethics, Privacy and Security

Ethics

Insights concerning specific demographics, particularly based on ethnicity, have the potential to reinforce biases about those demographics. For example, identifying a correlation between the proportion of Black residents in an area and lower income may contribute to existing stereotypes. Findings in this area must be presented with appropriate contextualisation, and an emphasis on systemic factors, rather than implying outcomes are a direct consequence of the demographic characteristics.

Privacy concerns

Every variable we use is a tract-level count, rate, or flag (e.g. PovertyRate, lablack1share, HUNVFlag) – there are no individual-level records (names, addresses, or household-level identifiers) anywhere in the dataset, so re-identification of a specific person is not a realistic risk.

A handful of tracts have a very low population count, and theoretically could be traced back to individuals. Whenever these tracts are used, they will be aggregated into larger groups such as by State or Urban/Rural status, removing any concerns of traceability.

Security

LRAM data is publicly available for personal use; steps do not need to be taken to secure the data itself.

Data and code for this project is stored in a Github repository. Devices that contain local copies of the code are password-protected, and the Github accounts are secured using SSH protocol. Analysis and findings can be stored securely in a private Github repository, only granting access to individuals working on the project; however, for the sake of accessibility for assessment, the repository for this project has been made public.

Exploratory Data Analysis

Exploring Low Access by Ethnicity, Income, and Poverty Rates

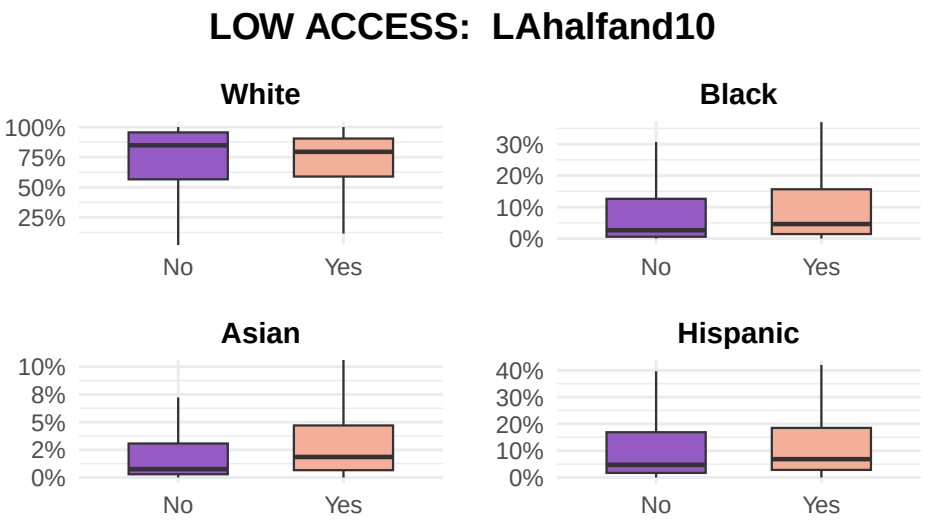


Figure 1.1: Percentages of Census Tract populations that are various ethnicities. Census Tracts are split by Low access flags, measured at a 1/2 mile distance for Urban and 10 miles for Rural. Ethnicities are shown on different scales for visibility

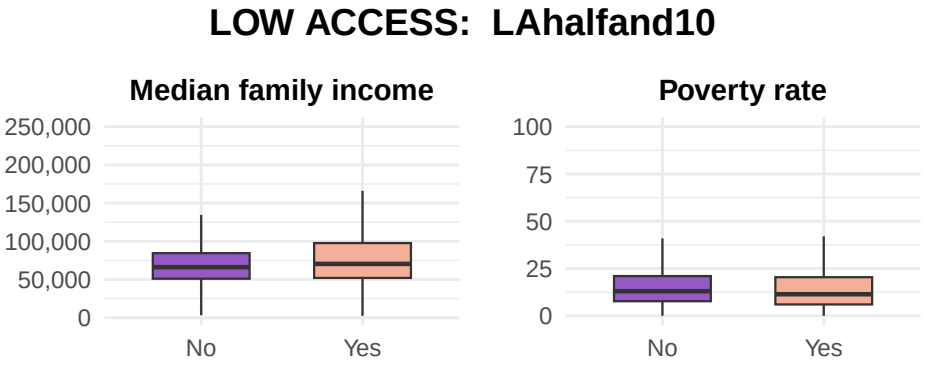


Figure 1.2: Median Family Income and Poverty Rates of Census Tracts, split by Low access flags, measured at a 1/2 mile distance for Urban and 10 miles for Rural.

Ethnicity shows minimal differences in medians when separating Census Tracts by Low Access status at 1/2 and 10 miles.

The difference in income and poverty rate between Census Tracts by Low Access status is similarly weak: tracts flagged as low access actually have slightly *higher* median income ($70.4k vs $66.1k) and slightly *lower* poverty (11.4% vs 13.0%) than non-flagged tracts.

Exploring differences in Urban and Rural Census Tracts

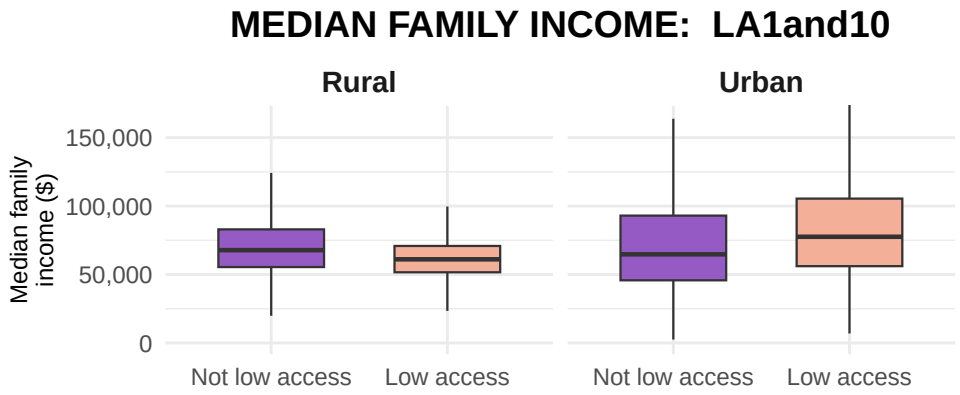


Figure 2: Median family income for Urban and Rural census Tracts, where Census Tracts are further split by **Low access Low income** flags, measured at a 1/2 mile distance for Urban and 10 miles for Rural.

In rural tracts, we see that low access tracts exhibit lower median family incomes.

However, in Urban tracts this effect is reversed. This reversed effect may be caused by densely populated affluent suburban neighbourhoods that do not have a local grocery store. This may explain the lack of noticeable difference in Median Income when grouping Urban and Rural Census Tracts together in Figure 1.2.

Exploring Low Income and Low Access (LILA) flags by Ethnicity

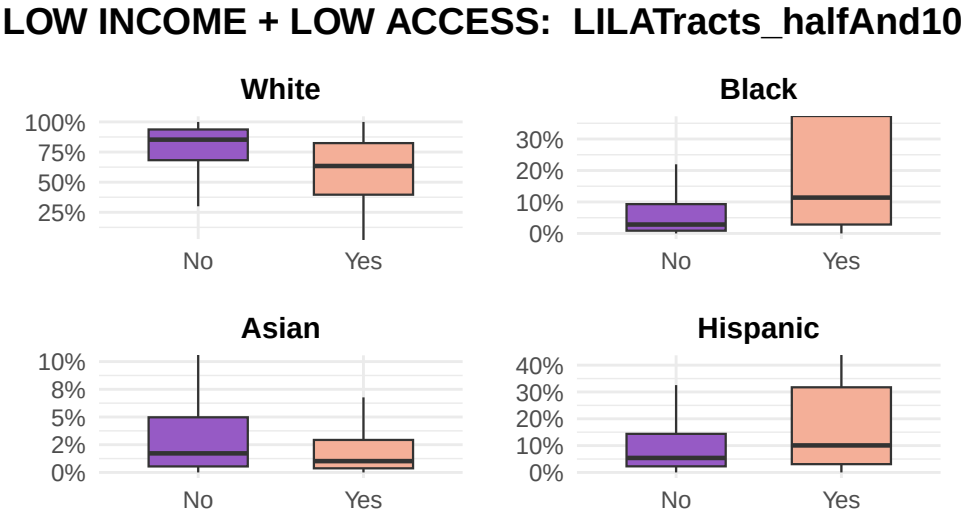


Figure 3: Percentages of Census Tract populations that are various ethnicities. Census Tracts are split by combined **Low access** and **Low income** flags, measured at a 1/2 mile distance for Urban and 10 miles for Rural. Ethnicities are shown on different scales for visibility

Figure 3 demonstrates there are different relationships between various and the Census Tract's status for Low income and low access (measured at 1/2 mile for Urban and 1 mile for Rural).

The largest ethnicity-based difference between the two statuses is seen in the percentage of the population that is White: the median percentage for census tracts that are flagged is around 60%, whereas the median percentage for census tracts that are not flagged is closer to 90%.

In contrast, the median percentage of Black individuals in the population for census tracts that are flagged is around 10%, whereas the median percentage of Black individuals in the population for census tracts that are not flagged is less than 5%. This effect is in the opposite direction to the effect of the percentage of White individuals in the population.

We see smaller differences in the percentage of the populations that are Asian and Hispanic between the two statuses, with a lower percentage of Asian individuals in Census Tracts considered low income and low access compared to census tracts without this status, and a higher percentage of Hispanic individuals in Census Tracts considered low income and low access.

Differences emerge in demographic percentages in Figure 3 that are not present in Figure 1.1, implying that the observed effect may be largely explained by the low income discriminating feature.

Detailed Data Analysis

Analysis techniques

A **logistic regression** model was fit. This model type is suitable for predicting information with two classes, such as whether a Census Tract is considered to have or not have a significant Low Income and Low Access population.

Statistically significant coefficients of a logistic regression model model can describe in further detail the influence that different demographic information has on the status of the Census Tract. This is in contrast to other classification methods such as neural networks that and are difficult to interpret relationships from, and are slower to compute.

Data manipulation, and addressing scope/outliers

For demographic explanatory terms (ethnicity and age-based continuous variables), total counts were converted to a logarithmically scaled percentage using the following equation: `log((<demographic population count>/Pop2010)*100 + 1)`.

Converting to percentages prevents the model fitting inadvertently to the population size. Many of the explanatory variables are skewed towards 0; taking the logarithm resulted in a better fitted model (AIC=65831, compared to AIC=69886 of a similar model fitted on the percentages without converting to the logarithmic scale).

Census Tracts from the states of Hawaii and Alaska were removed. These data points were frequently found to be outliers, as their ethnicity profiles are skewed very differently to other states, due to geographical separation and historical differences Outcomes are thus only applicable to states in the main land mass of North America.

Census Tracts where the total population was less than 150 were removed, which excluded 253 Census Tracts (167 Urban, 86 Rural). For Census Tracts with a population count at 150 or less, only 1 or 2 people are required to change the demographic by 1%. This also makes results difficult to interpret for these Census Tracts. Given the volume removed in comparison to the total records in Urban and Rural, this is not expected to bias the results.

Lastly, three remaining rural Census Tracts were removed as outliers; one from Arizona and two from Texas. Their demographic patterns were distinct from other Census Tracts (for example, in the Census Tract from Arizona, 5444 out of 5628 individuals were Native American or Alaskan).

Final model design

The response variable is **Low Income and Low Access at 1/2 and 10 miles**, where Y=1 corresponds to being Low Income and Low Access.

Ethnicity percentages were included as explanatory terms, measured on a logarithmic scale (see **data manipulation techniques**). **% White** was treated as the reference level and not included.

The interactions between whether the census tract was Urban or Rural (`Urban_f`), and a given ethnicity percentage, were included where found to be statistically significant (p-value < 0.001), with Urban as the reference level.

% Kids was included, and **% Seniors** excluded, as they are mutually exclusive categories.

Other models were fit that included interactions between **% Kids** and ethnicities such as **% Black**. When only a selection of interactions with ethnicities were considered, the coefficients were found to be insignificant (for many, the p-value was > 0.05). A model was fit that incorporated all possible interaction terms; all of them were significant (p-value < 0.001) but model interpretation was compromised.

Model fit was analysed by comparing explanatory terms to fitted values. When a significant coefficient for the interaction with Urban/Rural was found, comparisons were made on each type separately. These were found to be **linear**, confirming the model design is appropriate for the shape of the data.

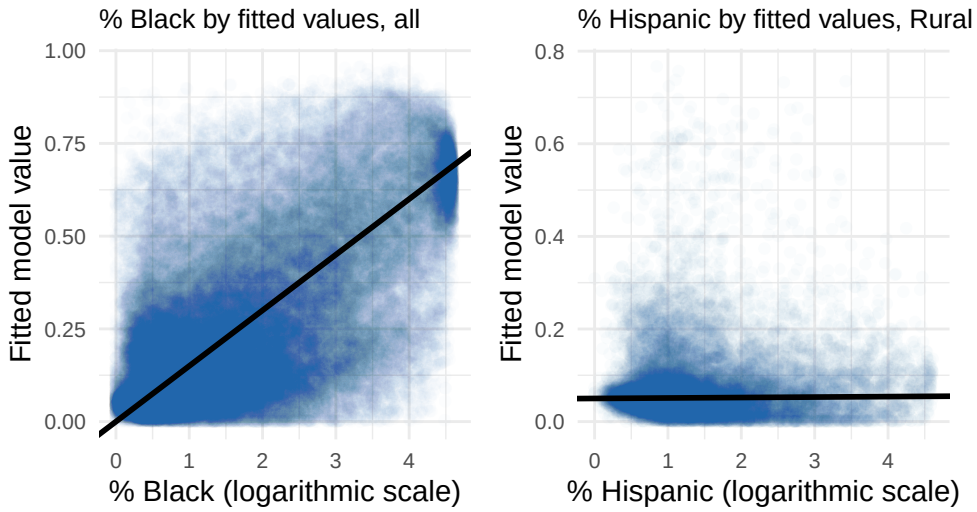


Figure 3: Correlations between explanatory terms and model fitted values, separating Urban and Rural where applicable based on model design.

Interpreting the model

All tables present direct effects on the odds (i.e. the exponent of the raw coefficient) of % changes on a **logarithmic scale**. This means that increases from lower values are more impactful than increases from higher values: the effect of an increase in a demographic from 1% to 2% is of equal magnitude than an increase in a demographic from 10% to 20%.

In the following discussion a "logarithmic unit increase" refers to an increase in the percentage that equates to 1 on the logarithmic scale.

Table 1: Demographic effects on odds in Urban Census Tracts

	Effect on Odds (log scale)	Lower CI	Upper CI	P-value
% Kids	0.77	0.73	0.81	<2e-16
% Asian	0.37	0.36	0.38	<2e-16
% Hispanic	0.61	0.58	0.64	<2e-16
% Hawaiian/Other Pacific Islander	2.15	1.93	2.39	<2e-16
% Native American/Alaska Native	1.51	1.42	1.60	<2e-16
% Multiracial/Other	3.36	3.12	3.62	<2e-16
% Black	1.50	1.48	1.52	<2e-16

Table 2: Overall effect of demographics on odds in Rural Census Tracts (equal to Urban effect times additional Rural effect)

	Effect
% Asian (Rural)	0.09
% Hispanic (Rural)	1.27
% Native American/Alaska Native (Rural)	1.99
% Multiracial/Other (Rural)	0.81

The odds of a Rural Census Tracts being Low Income, Low Access is 55% less than the odds for Urban Census Tracts, as seen in Table 3 in the Appendix (95% CI (0.38, 0.52), p-value $<$ 2e-16). This may be a consequence of inequities between Population Density types, but it may also be a consequence of our choice of distances for Urban and Rural areas in our response variable.

Black population and Low Income, Low Access

There is a positive relationship with **% Black** and the odds of being Low Income, Low Access: Table 1 shows for each logarithmic unit increase of % Black, the odds of being Low Income, Low Access for Urban Census Tracts increase by 50% (95% CI (1.48, 1.52), p-value $<$ 2e-16). No statistically significant difference was found in Rural areas, and so this term was not included in the final model.

Figure 4 shows that the percentage of a Census Tract that is Black frequently sits between 0 and 10%. For a Census Tract with the median total population size of approximately 4000, an 1% increase would equate to 40 people. Note that an increase from 10% to approximately 29% corresponds to a 1 unit increase on our logarithmically transformed scale ($\log(10+1)+1 \approx \log(28.9+1)$), a difference of 19%. A difference of 760 Black people could represent a 50% increase of the odds of being Low Income, Low Access. It is important to note this is does not imply causation (if 760 Black people relocated this will not necessarily change the status of the Census Tracts) but rather indicates a clear lacking in the opportunities and infrastructure available in areas with a more prevalent Black population.

Worse outcomes for smallest ethnic groups

Similar in size to % Black, a positive relationship was found with **% Native American/Alaska Native** in Urban areas (51% increase, 95% CI (1.42, 1.60), p-value $<$ 2e-16). In Rural areas, the odds are nearly 100% higher per logarithmic unit increase of Native American/Alaska Natives (additional multiplicative effect of 32% in Rural areas, 95% CI (1.21, 1.45), p-value = 6.32e-10). Census Tracts more frequently have a lower percentage of Native American/Alaska Native individuals than Black individuals (Figure 4); given the logarithmic scale the model was fitted to we can say that these smaller ethnic groups are disproportionately impacted.

Tables 1 and 2 also show the largest positive relationships are in **% Hawaiian/Other Pacific Islander** in both Urban and Rural areas, and for **% Multiracial/Other** specifically in Urban areas. These ethnicities also tend to represent smaller percentages of the population.

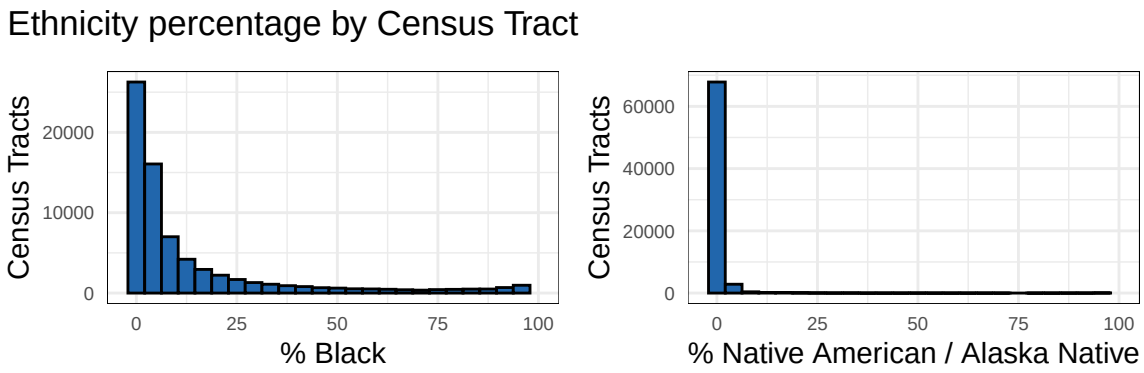


Figure 4: Distribution of all Census Tracts used in the logistic regression model, by % measurements of their Black and Native American / Alaska Native populations.

Mixed relationships with Hispanic and Multiracial/Other, Urban vs Rural

There is a negative relationship with **% Hispanic** in Urban areas: for each logarithmic unit increase in %Hispanic in Urban areas, the odds **decrease** by 39% (95% CI (0.58, 0.64), p-value $<$ 2e-16). However, an inverse effect is observed in Rural areas, where for each logarithmic unit increase the odds **increase** by 27% (additional in Rural areas of 2.09, 95% CI (1.80, 2.42), p-value $<$ 2e-16). A similar inversion between Urban and Rural Census Tracts is observed for **%Multiracial/Other**.

This may indicate better outcomes for the Hispanic population in Urban areas compared to Rural, in distance to food, or general income. However, it may also be a consequence of confounding variables not modelled, for example location-based patterns in % Hispanic that differ between Urban and Rural. This could be analysed by more closely examining the median income or percentage of the Hispanic population that is Low Access in Urban and Rural Census Tracts separately. However, this data is unavailable, or significant proportions of it are missing, in the LRAM dataset. More detailed qualitative or quantitative analysis that pulls data from other sources would be needed to make stronger conclusions here.

Conclusions and Recommendations

Note that a "worse" outcome refers to higher odds that a Census Tract is flagged as Low Income and Low Access.

1. Slightly better outcomes were observed in Census Tracts with a larger percentage of children. However, no conclusions could be drawn on the interaction of children and different ethnicities. Note that no information was available on the breakdown of the percentage of children that belonged to different ethnicities .
2. Better outcomes were observed in Census Tracts with larger Asian populations in both Urban and Rural, while worse outcomes were observed in Census Tracts with larger Black populations.
3. The smallest minority groups in the United States of America, including Hawaiian/Other Pacific Islander, and Native American, have a disproportionately strong relationship with worse outcomes.
4. Distinctions between Urban and Rural areas are clear, and impacts vary for different ethnic groups.

Changes and targeted initiatives should prioritise or strongly consider the impact on areas with a stronger presence of Native American, Alaskan Native, Hawaiian, and other Pacific Islander groups, to prevent exacerbating existing inequalities these communities face.

Black communities in Urban and Rural areas have been similarly affected and may both benefit from targeted initiatives designed to address these inequalities. However, in general, impacts on communities in Urban and Rural should be considered separately.

These findings cannot be applied to the States of Alaska and Hawaii, and not to Census Tracts where the population is smaller than 150. Further research focussing on these areas may be beneficial for further understanding.

Further research may also include gathering data on ages within different demographics (for example, how many Black children are in an area compared to how many White children) which may improve understanding of relationships with other compounding difficulties such as the capacity to support non-working family members.

In order to uphold ethical standards, findings should be considered alongside quantitative knowledge of these communities, their opinions and cultures, when planning action from these findings.

References

(1) Economic Research Service, U.S. Department of Agriculture. (2019). *Food Access Research Atlas: Large Retailer Access Map (LRAM) documentation.* https://www.ers.usda.gov/data-products/food-access-research-atlas/documentation

(2) U.S. Department of Agriculture, Economic Research Service. (2026, July 27). *Food Access Research Atlas – Large Retailer Access Map Reference Guide* https://www.ers.usda.gov/data-products/food-access-research-atlas/documentation/large-retailer-access-map-reference-guide

A note on AI usage for the Detailed Data Analysis

Artificial Intelligence, specifically ChatGPT, was used to help find typos in written text, suggest wording improvements, to fix / alter code that generates more complex plots, and to help with adjusting general RMarkdown settings (e.g. suppressing warnings, setting plot sizes). The analysis itself was written manually, and all suggested code changes were reviewed for accuracy. No university material was provided to the AI for these purposes.

Appendix

Table 3: Additional Demographic effects on odds in Rural Census Tracts

	Effect on Odds (log scale)	Lower CI	Upper CI	P-value
Baseline difference compared to Urban	0.45	0.38	0.52	<2e-16
% Asian (Rural)	0.24	0.18	0.31	<2e-16
% Hispanic (Rural)	2.09	1.80	2.42	<2e-16
% Native American/Alaska Native (Rural)	1.32	1.21	1.45	6.32e-10
% Multiracial/Other (Rural)	0.24	0.19	0.30	<2e-16